

CSA0904 – PROGRAMMING FOR JAVA

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COLLECTION OVER ALL EXCERCISE

1

The screenshot shows the SIMATS Online Programming Lab interface. The code editor contains a Java program that uses a HashMap to store department names and their corresponding employee lists. The program iterates through the HashMap and prints the department name followed by the list of employees.

```
3 class Main {
4     public static void main(String[] args) {
5
6         HashMap<String, ArrayList<String>> emp = new HashMap<>();
7
8         ArrayList<String> it = new ArrayList<>();
9         it.add("101 Ram 50000");
10        it.add("102 Ravi 60000");
11
12        ArrayList<String> hr = new ArrayList<>();
13        hr.add("103 Priya 45000");
14        hr.add("104 Anu 55000");
15
16        emp.put("IT", it);
17        emp.put("HR", hr);
18
19        Iterator<String> d = emp.keySet().iterator();
20
21        while (d.hasNext()) {
22            String dept = d.next();
23            System.out.println("Department: " + dept);
24
25            ArrayList<String> list = emp.get(dept);
26
27            for (String e : list)
28                System.out.println(e);
29        }
30    }
31 }
```

The output of the program is displayed on the right side of the interface:

```
Output
Department: HR
103 Priya 45000
104 Anu 55000
Department: IT
101 Ram 50000
102 Ravi 60000
```

2.

The screenshot shows the SIMATS Online Programming Lab interface. The code editor contains a Java program that uses a Stack to simulate a web browser's back and forward functionality. The program pushes URLs onto the stack, prints the current page, and then pops the stack to print the back history. It also prints the forward history by iterating through the stack.

```
1 import java.util.*;
2
3 class Main {
4     public static void main(String[] args) {
5         Stack<String> back = new Stack<>();
6         Stack<String> forward = new Stack<>();
7         back.push("google.com");
8         back.push("youtube.com");
9         back.push("github.com");
10        System.out.println("Current Page: " + back.peek());
11        forward.push(back.pop());
12        System.out.println("After Back: " + back.peek());
13        back.push(forward.pop());
14        System.out.println("After Forward: " + back.peek());
15        back.push("openai.com");
16        forward.clear();
17        System.out.println("\nBack History:");
18        Iterator<String> it = back.iterator();
19        while (it.hasNext())
20            System.out.println(it.next());
21        System.out.println("\nForward History:");
22        Iterator<String> it2 = forward.iterator();
23        while (it2.hasNext())
24            System.out.println(it2.next());
25    }
26 }
```

The output of the program is displayed on the right side of the interface:

```
Output
Current Page: github.com
After Back: youtube.com
After Forward: github.com
Back History:
google.com
```

3.RAILWAY PARCEL DISPATCH



SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING



Dharshan A N
192525370RA

4/4 answered

Your Code

```

1 import java.util.*;
2
3 class Main {
4     public static void main(String[] args) {
5         ArrayList<HashMap<String, Object>> list = new ArrayList<>();
6         HashMap<String, Object> s1 = new HashMap<>();
7         s1.put("Name", "John"); s1.put("Math", 80); s1.put("Java", 70);
8         HashMap<String, Object> s2 = new HashMap<>();
9         s2.put("Name", "Ram"); s2.put("Math", 30); s2.put("Java", 20);
10        HashMap<String, Object> s3 = new HashMap<>();
11        s3.put("Name", "Sam"); s3.put("Math", 90); s3.put("Java", 95);
12        list.add(s1);
13        list.add(s2);
14        list.add(s3);
15        for(HashMap<String, Object> s : list){
16            int m = (Integer)s.get("Math");
17            int j = (Integer)s.get("Java");
18            int avg = (m+j)/2;
19            if(m>=40 || j>=40)
20                System.out.println(s.get("Name")+" Average = "+avg);
21            else
22                System.out.println(s.get("Name")+" Failed");
23        }
24    }
25 }
                
```




Input

Output

```

John Average = 75
Ram Failed
Sam Average = 92
                
```

4.

The image is a screenshot of a web browser displaying the SIMATS Online Programming Lab. The browser's address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=683`. The page has a dark blue header with the SIMATS ENGINEERING logo on the left, the title "SIMATS Online Programming Lab" in the center, and a user profile "Dharshan A N" with ID "192525370RA" on the right. The main content area is divided into two panels. The left panel is a code editor with line numbers 6 to 34, containing the following Java code:

```
6      HashSet<String> books = new HashSet<>();
7
8      books.add("101 Java James 2022");
9      books.add("102 Python Guido 2023");
10     books.add("103 C++ James 2021");
11
12     ArrayList<String> list = new ArrayList<>(books);
13
14     Collections.sort(list, Collections.reverseOrder());
15
16     System.out.println("Books:");
17     Iterator<String> it = list.iterator();
18
19     while (it.hasNext())
20         System.out.println(it.next());
21
22     System.out.println("\nBooks by James:");
23     for (String b : list) {
24         if (b.contains("James"))
25             System.out.println(b);
26     }
27
28     System.out.println("\nBook with ISBN 102 Removed (Displayed On");
29     for (String b : list) {
30         if (!b.startsWith("102"))
31             System.out.println(b);
32     }
33
34 )
```

The right panel is an output window titled "Output" showing the execution results:

```
Books:
103 C++ James 2021
102 Python Guido 2023
101 Java James 2022

Books by James:
```

The bottom of the screen shows a Windows taskbar with various application icons, a system clock displaying "11:18" and "12-08-2026", and a weather widget showing "32°C Mostly sunny".